

LibeRary reference manual

Pharmacometric Model Repository for the LibeR Ecosystem

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88 help topics

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Adjudicate discrepancies using a third configured LLM role

Description

Adjudicate discrepancies using a third configured LLM role

Usage

```
ingest_adjudicate_extractions(  
  metadata,  
  text,  
  vision,  
  comparison,  
  bundle,  
  cfg = NULL  
)
```

Arguments

metadata	Publication metadata.
text	Text-lane result.
vision	Vision-lane result.
comparison	Field comparison.
bundle	Document bundle.
cfg	LibeRary configuration.

Value

Adjudication result, or an unavailable result with an error.

Independently assess an extracted model against its evidence

Description

Independently assess an extracted model against its evidence

Usage

```
ingest_assess_model(  
    metadata,  
    extraction,  
    cfg = NULL,  
    full_text = "",  
    supplement_text = ""  
)
```

Arguments

metadata	Publication metadata.
extraction	Parsed extraction.
cfg	Ingest configuration.
full_text	Publication text.
supplement_text	Supplement text.

Value

Assessment and audit data.

Classify acquisition route for a literature record

Description

Classify acquisition route for a literature record

Usage

```
ingest_classify_entry(metadata, cfg, triage = NULL)
```

Arguments

metadata	Record from <code>ingest_entrez_fetch_metadata()</code> .
cfg	Ingest config.
triage	Optional result from <code>ingest_triage_abstract()</code> . When supplied, its probability tier determines whether the record belongs to the first processing pass or the retained low-probability backlog.

Value

Manifest row list.

Compare text- and vision-lane model claims field by field

Description

Compare text- and vision-lane model claims field by field

Usage

```
ingest_compare_extractions(text, vision, tolerance = 0.02)
```

Arguments

text	Text-lane result.
vision	Vision-lane result.
tolerance	Relative tolerance for numeric claims.

Value

Reconciliation report.

Default PubMed query for pharmacometric literature

Description

Default PubMed query for pharmacometric literature

Usage

```
ingest_default_query()
```

Value

Character scalar PubMed query string.

Run a staged evidence-led model investigation

Description

The article is mapped into searchable chunks, investigated by pharmacometric topic, challenged by a skeptical reviewer, optionally re-searched for gaps, checked deterministically, and only then synthesized into the standard LibeRary extraction schema. Every stage is content-addressed and resumable.

Usage

```
ingest_deliberative_extract(  
  metadata,  
  bundle,  
  cfg = NULL,  
  progress = NULL,  
  force = FALSE,  
  chat = library_llm_chat  
)
```

Arguments

metadata	Publication metadata.
bundle	Document bundle or bundle path.
cfg	LibeRary configuration.
progress	Optional callback accepting value, message, and stage.
force	Ignore reusable stage caches.
chat	Chat function, primarily for deterministic testing.

Value

A text-lane-compatible result with an evidence-led audit.

Discover PubMed entries and classify acquisition routes

Description

Runs E-utilities search (rate-limited), fetches metadata, resolves OA sources, downloads open-access PDFs immediately, and returns manifest rows.

Usage

```
ingest_discover(  
  query = DEFAULT_PM_QUERY,  
  limit = 20L,  
  cfg = NULL,  
  triage = TRUE,  
  download_oa = TRUE,  
  log = NULL,  
  progress = NULL  
)
```

Arguments

query	PubMed query. Defaults to <code>ingest_default_query()</code> .
limit	Maximum PMIDs to process.
cfg	Config from <code>ingest_load_config()</code> or path to YAML.
triage	If TRUE, assess model probability from title and abstract before acquisition routing. High and intermediate tiers enter the first pass; low-probability records are retained in a separate backlog.
download_oa	If TRUE, download oa_auto PDFs during discovery.

log

Optional logger function(msg, level).

progress

Optional function(value, message, step, total).

Value

List with pmids, entries (manifest rows), metadata, summary.

Check whether the configured Docling command is available

Description

Check whether the configured Docling command is available

Usage

```
ingest_docling_available(cfg = NULL)
```

Arguments

cfg	LibeRary configuration.
-----	-------------------------

Value

A list containing available, executable, and version.

Parse a PDF with Docling's standard document pipeline

Description

This lane deliberately uses Docling's standard parser rather than its VLM pipeline, keeping it independent from the raw-page vision extraction lane. A test or deployment runner can be supplied as options(`LibeRary.docling_runner = function(executable, args, stdout, stderr) ...`).

Usage

```
ingest_docling_parse(pdf_path, output_dir, cfg = NULL, force = FALSE)
```

Arguments

pdf_path	PDF to convert.
output_dir	Conversion output directory.
cfg	LibeRary configuration.
force	Re-run when converted output already exists.

Value

Conversion manifest.

Build a canonical, content-addressed document bundle

Description

The bundle retains the source PDF, standard-parser outputs, selected raw page images, hashes, parser provenance, and fallback status. It is safe to resume: an unchanged PDF reuses its existing bundle.

Usage

```
ingest_document_bundle(metadata, pdf_path, cfg = NULL, force = FALSE)
```

Arguments

metadata	Publication metadata.
pdf_path	Full-text PDF.
cfg	LibeRary configuration.
force	Rebuild an existing bundle.

Value

Bundle manifest.

Download a PDF to inbox for a PMID

Description

Download a PDF to inbox for a PMID

Usage

```
ingest_download_pdf(pmid, url, cfg, dest_name = "article.pdf")
```

Arguments

pmid	PMID string.
url	URL to download.
cfg	Ingest config.
dest_name	Destination filename inside inbox/<pmid>/.

Value

List with success, path, final_url, error.

Run independent text and vision extraction with automated reconciliation

Description

Consistent records are labelled `machine_consistent`. Conflicts are sent to the adjudication role and labelled `machine_adjudicated` unless a major field remains unresolved. Machine agreement is deliberately not called human validation.

Usage

```
ingest_dual_extract(  
  metadata,  
  bundle,  
  cfg = NULL,  
  adjudicate = TRUE,  
  progress = NULL  
)
```

Arguments

metadata	Publication metadata.
bundle	Document bundle or path to one.
cfg	LibeRary configuration.
adjudicate	Automatically adjudicate discrepancies.
progress	Optional callback for progress within this article, accepting value, message, and stage.

Value

Reconciled extraction, status, lane outputs, and audit data.

Count PubMed hits for a query (no metadata fetch)

Description

Count PubMed hits for a query (no metadata fetch)

Usage

```
ingest_entrez_count(query, cfg)
```

Arguments

query	PubMed query string.
-------	----------------------

cfg	Ingest config.
-----	----------------

Value

Integer total hit count reported by NCBI.

Fetch PubMed metadata for PMIDs

Description

Fetch PubMed metadata for PMIDs

Usage

```
ingest_entrez_fetch_metadata(pmids, cfg, use_cache = TRUE)
```

Arguments

pmids	Character vector of PMIDs.
cfg	Ingest config.
use_cache	If TRUE, read/write per-PMID JSON cache.

Value

A list of metadata records (named by PMID).

Search PubMed via E-utilities

Description

Search PubMed via E-utilities

Usage

```
ingest_entrez_search(query, retmax = 20L, cfg)
```

Arguments

query	PubMed query string.
-------	----------------------

retmax	Maximum PMIDs to return.
--------	--------------------------

cfg	Ingest config from <code>ingest_load_config()</code> .
-----	--

Value

Character vector of PMIDs.

Europe PMC lookup for full-text PDF hints

Description

Europe PMC lookup for full-text PDF hints

Usage

```
ingest_europe_pmc_lookup(pmid = "", doi = "", cfg)
```

Arguments

pmid	PubMed ID.
doi	Optional DOI.
cfg	Ingest config.

Value

List with pdf_url, is_open_access, pmcid.

Batch extract, assess and publish discovered publications

Description

Batch extract, assess and publish discovered publications

Usage

```
ingest_extract_batch(  
  manifest,  
  cfg = NULL,  
  limit = 10L,  
  use_ollama = TRUE,  
  assess = TRUE,  
  resume = TRUE,  
  overwrite = TRUE,  
  log = NULL,  
  progress = NULL  
)
```

Arguments

manifest	Discover manifest path or data frame.
cfg	Ingest configuration.
limit	Maximum records.
use_ollama	Legacy switch; FALSE creates metadata stubs.
assess	Run the independently configured assessment role.
resume	Skip PMIDs already present in the target catalogue.

overwrite

Update existing entries when resume=FALSE.

log

Optional logger.

progress

Optional progress callback.

Value

Batch results and provider summary.

Run model extraction for one publication

Description

Run model extraction for one publication

Usage

```
ingest_extract_model(  
  metadata,  
  cfg = NULL,  
  pdf_path = NULL,  
  assess = TRUE,  
  full_text = NULL,  
  supplement_text = NULL  
)
```

Arguments

metadata	PubMed metadata.
cfg	Ingest configuration.
pdf_path	Optional publication PDF.
assess	Run a separate evidence assessment after extraction.
full_text	Optional pre-extracted article text, used by remote workers.
supplement_text	Optional pre-extracted supplement text.

Value

Extraction, assessment and immutable audit metadata.

Extract a model from the standard-parser text lane

Description

Extract a model from the standard-parser text lane

Usage

```
ingest_extract_text_lane(metadata, bundle, cfg = NULL, progress = NULL)
```

Arguments

metadata	Publication metadata.
bundle	Document bundle.
cfg	LibeRary configuration.
progress	Optional callback accepting value, message, and stage.

Value

Lane result with extraction and immutable audit data.

Extract a model directly from original PDF page images

Description

Extract a model directly from original PDF page images

Usage

```
ingest_extract_vision_lane(metadata, bundle, cfg = NULL)
```

Arguments

metadata	Publication metadata.
bundle	Document bundle.
cfg	LibeRary configuration.

Value

Lane result with extraction and immutable audit data.

Fetch pending institutional (paywalled) PDFs

Description

Attempts direct PDF URLs when known, otherwise follows DOI redirects on the institutional/VPN network. Optional chromote fallback for publisher pages that require JavaScript.

Usage

```
ingest_fetch_institutional(  
  manifest,  
  cfg = NULL,  
  classes = "needs_institutional",  
  tiers = NULL,  
  use_chromote_fallback = NULL,  
  log = NULL,  
  progress = NULL  
)
```

Arguments

manifest	Path to manifest CSV or data frame.
cfg	Ingest config.
classes	Character vector of acquisition_class values to fetch. Default: "needs_institutional".
tiers	Probability tiers to fetch. Defaults to the configured first pass (high and intermediate). Use "low" to process the retained backlog later.
use_chromote_fallback	Use chromote when direct download fails.

log

Optional logger function.

progress

Optional progress callback.

Value

List with results, summary, failures, log_path.

Detect supplement control streams in a directory

Description

Detect supplement control streams in a directory

Usage

```
ingest_find_supplements(dir)
```

Arguments

dir Directory to scan (for example, `inbox/<pmid>`).

Value

Character vector of matching file paths.

Generate standalone institutional fetch R script

Description

Writes a script intended to run on a VPN-enabled work laptop. The script loads LibeRary and calls `ingest_fetch_institutional()`.

Usage

```
ingest_generate_fetch_script(  
  manifest_path,  
  output_path = "library_fetch_institutional.R",  
  config_path = NULL  
)
```

Arguments

manifest_path	Path to manifest CSV.
output_path	Path for generated R script.
config_path	Optional YAML config path embedded in script.

Background worker for LibeRary GUI jobs

Description

Background worker for LibeRary GUI jobs

Usage

```
ingest_gui_worker(params)
```

Arguments

params	List with job settings (see Shiny app).
--------	---

Value

List with ok and result or error.

Find most recent discover manifest

Description

Find most recent discover manifest

Usage

```
ingest_latest_manifest(cfg)
```

Arguments

cfg	Ingest config list.
-----	---------------------

Value

File path or "".

Load ingest configuration

Description

Reads YAML config when the **yaml** package is installed; otherwise uses defaults merged with environment variables.

Usage

```
ingest_load_config(path = NULL)
```

Arguments

path Path to YAML config file. If missing, uses env vars only.

Value

A normalized config list.

Create a file-backed logger for GUI / background jobs

Description

Create a file-backed logger for GUI / background jobs

Usage

```
ingest_make_logger(log_path)
```

Arguments

log_path	Path to append log lines.
----------	---------------------------

Value

Function `function(msg, level = "INFO")`.

Map extracted evidence to a reviewable NONMEM control stream

Description

Only established LibeRation templates (ADVAN1-4/11/12) are generated. Any missing structural or parameter evidence is explicitly recorded in the mapping attribute and the resulting catalogue entry cannot be promoted to validated status without review.

Usage

```
ingest_map_to_ctl(extraction, metadata = NULL)
```

Arguments

extraction	Parsed extraction list.
metadata	Optional publication metadata.

Value

Character control stream with a mapping attribute.

Legacy Ollama provider wrappers

Description

Compatibility functions for scripts written against the original ingest prototype. New code should use `library_llm_available()` and `library_llm_chat()`.

Usage

```
ingest_ollama_available(cfg)

ingest_ollama_chat(messages, cfg, format = NULL)
```

Arguments

<code>cfg</code>	LibeRary configuration.
<code>messages</code>	Chat messages.
<code>format</code>	Optional structured-output schema.

Value

`ingest_ollama_available()` returns a logical scalar; `ingest_ollama_chat()` returns the response content as text.

Parse JSON from an LLM response

Description

Parse JSON from an LLM response

Usage

```
ingest_parse_llm_json(text, schema = NULL)
```

Arguments

text Raw response, optionally fenced or surrounded by short prose.

schema Optional JSON-schema-like list used to validate the decoded response before it reaches semantic model code.

Value

Parsed list.

Extract text from a PDF file

Description

Uses **pdftools** when available.

Usage

```
ingest_pdf_text(path, max_chars = 120000L)
```

Arguments

path	Path to PDF.
------	--------------

max_chars	Truncate extracted text (LLM context limit).
-----------	--

Value

Character scalar of extracted text.

Check PMC open-access availability and PDF link

Description

Check PMC open-access availability and PDF link

Usage

```
ingest_pmc_oa_lookup(pmcid, cfg)
```

Arguments

pmcid	PMC ID with or without prefix.
-------	--------------------------------

cfg	Ingest config.
-----	----------------

Value

List with is_oa, pdf_url.

Build deterministic PMC PDF URL

Description

Build deterministic PMC PDF URL

Usage

```
ingest_pmc_pdf_url(pmcid)
```

Arguments

<code>pmcid</code>	PMC ID with or without PMC prefix.
--------------------	------------------------------------

Value

URL string or empty if no PMCID.

Process a triaged literature batch into reconciled catalogue entries

Description

By default only High and Intermediate probability records are processed. Low-probability records remain in the manifest and can be processed later by setting `tiers = "low"`. Each PDF is converted to a resumable document bundle, independently extracted through text and vision lanes, compared field by field, and automatically adjudicated when necessary.

Usage

```
ingest_process_batch(  
  manifest,  
  cfg = NULL,  
  tiers = NULL,  
  limit = Inf,  
  resume = TRUE,  
  adjudicate = TRUE,  
  overwrite = TRUE,  
  continue_on_error = TRUE,  
  log = NULL,  
  progress = NULL,  
  article_progress = NULL  
)
```

Arguments

<code>manifest</code>	Discovery manifest path or data frame.
<code>cfg</code>	LibeRary configuration.
<code>tiers</code>	Triage tiers to process.
<code>limit</code>	Maximum records to process.

<code>resume</code>	Reuse terminal decisions and existing catalogue entries.
<code>adjudicate</code>	Run automatic third-model adjudication.
<code>overwrite</code>	Update an existing catalogue entry.
<code>continue_on_error</code>	Continue remaining articles after an article error.
<code>log</code>	Optional logger.
<code>progress</code>	Optional progress callback.
<code>article_progress</code>	Optional callback for progress within the current PMID, accepting value, message, step, total, pmid, stage, and status.

Value

Batch results and summary.

Publish an extracted model to the catalogue

Description

Publish an extracted model to the catalogue

Usage

```
ingest_publish_catalog_entry(  
    metadata,  
    extraction,  
    cfg = NULL,  
    status = "stub",  
    raw_llm = NULL,  
    library_id = NULL,  
    assessment = NULL,  
    overwrite = TRUE,  
    document_bundle = NULL  
)
```

Arguments

metadata	Publication metadata.
extraction	Extraction list.
cfg	Ingest configuration.
status	Initial workflow status.
raw_llm	Optional immutable LLM audit object.
library_id	Optional stable entry id.

assessment

Optional independent evidence assessment.

overwrite

Update an existing entry and archive its previous version.

document_bundle

Optional canonical document-bundle manifest.

Value

Published paths and mapping details.

Read a canonical document bundle

Description

Read a canonical document bundle

Usage

```
ingest_read_document_bundle(path)
```

Arguments

path	Bundle directory or bundle.json path.
------	---------------------------------------

Value

Bundle manifest.

Read manifest CSV

Description

Read manifest CSV

Usage

```
ingest_read_manifest(path)
```

Arguments

path	CSV path.
------	-----------

Value

Data frame.

Read GUI progress state

Description

Read GUI progress state

Usage

```
ingest_read_progress(progress_path)
```

Arguments

progress_path	JSON file path.
---------------	-----------------

Value

List or NULL.

Render selected original PDF pages for the independent vision lane

Description

Render selected original PDF pages for the independent vision lane

Usage

```
ingest_render_pdf_pages(  
    pdf_path,  
    output_dir,  
    pages = NULL,  
    dpi = 140L,  
    max_pages = 12L  
)
```

Arguments

pdf_path	PDF path.
output_dir	Page image directory.
pages	One-based page numbers, or NULL for relevance-based selection.
dpi	Image resolution.
max_pages	Maximum automatically selected pages.

Value

Normalized paths to PNG page images.

Launch the LibeRary literature-ingestion application

Description

Provides PubMed discovery and abstract triage, full-text acquisition, Docling parsing, independent text/vision extraction, automated adjudication, reviewable catalogue publication, and live background-job logs.

Usage

```
ingest_shiny(host = "127.0.0.1", port = NULL, launch.browser = TRUE)

library_ingest_shiny(host = "127.0.0.1", port = NULL, launch.browser = TRUE)
```

Arguments

host	Host passed to shiny::runApp().
port	Port or NULL.
launch.browser	Open a browser.

Value

The Shiny application, invisibly.

Minimal metadata-only extraction

Description

Minimal metadata-only extraction

Usage

```
ingest_stub_extraction(metadata, reason = "abstract_only")
```

Arguments

metadata	PubMed metadata.
----------	------------------

reason	Reason for creating a stub.
--------	-----------------------------

Value

Extraction-like list requiring review.

Triage one publication from its title and abstract

Description

The LLM score is converted to a deterministic high, intermediate, or low tier using configuration thresholds. Low-tier records are deferred, never discarded.

Usage

```
ingest_triage_abstract(metadata, cfg = NULL, fallback_on_error = TRUE)
```

Arguments

metadata	PubMed metadata.
cfg	LibeRary configuration.
fallback_on_error	Use the conservative deterministic fallback when the configured triage model cannot be reached.

Value

Structured triage decision and audit data.

Triage a collection of PubMed metadata records

Description

Triage a collection of PubMed metadata records

Usage

```
ingest_triage_batch(metadata, cfg = NULL, log = NULL, progress = NULL)
```

Arguments

metadata	Named list of metadata records.
cfg	LibeRary configuration.
log	Optional logger.
progress	Optional progress callback.

Value

A named list of triage decisions.

Look up open-access locations via Unpaywall

Description

Look up open-access locations via Unpaywall

Usage

```
ingest_unpaywall_lookup(doi, cfg, use_cache = TRUE)
```

Arguments

doi	DOI string.
cfg	Ingest config.
use_cache	Use disk cache under cache/unpaywall.

Value

List with is_oa, url_for_pdf, url, host_type, or NULL on failure.

Validate and normalize LibeRary configuration

Description

Validate and normalize LibeRary configuration

Usage

```
ingest_validate_config(cfg)
```

Arguments

cfg	Configuration list.
-----	---------------------

Value

Normalized configuration.

Write institutional fetch manifest CSV

Description

Write institutional fetch manifest CSV

Usage

```
ingest_write_manifest(entries, path)
```

Arguments

entries	Data frame or list of manifest rows.
---------	--------------------------------------

path	Output CSV path.
------	------------------

Write GUI progress state

Description

Write GUI progress state

Usage

```
ingest_write_progress(  
  progress_path,  
  value,  
  message,  
  step = NA_integer_,  
  total = NA_integer_,  
  status = "running",  
  current = NULL  
)
```

Arguments

progress_path	JSON file path.
value	Progress fraction 0–1.
message	Status message.
step	Optional current step index.
total	Optional total steps.
status	One of idle, running, done, error, or cancelled.

current

Optional nested progress record for the current PMID. When omitted, an existing current-PMID record is retained.

Catalogue root directory

Description

The mutable catalogue lives in the user's application-data directory, so it survives upgrades and reinstalls. Packaged validated examples are seeded on first use.

`options(LibErary.catalog=)` or `LIBRARY_CATALOG` can select a shared or project-specific catalogue.

Usage

```
library_catalog_root(create = TRUE)
```

Arguments

<code>create</code>	Create and seed the catalogue when it does not exist.
---------------------	---

Value

Normalized directory path.

Upgrade-safe LibeRary configuration path

Description

Secrets are deliberately read from environment variables rather than saved in package files. The returned YAML path survives package upgrades.

Usage

```
library_config_path(create = FALSE)
```

Arguments

create	Create its parent directory.
--------	------------------------------

Value

Normalized configuration path.

Normalize reported dosing information

Description

Normalize reported dosing information

Usage

```
library_dosing_normalize(dose, route = NULL, source_locator = "")
```

Arguments

dose	Reported dose text or existing regimen list.
route	Optional normalized route.
source_locator	Evidence locator.

Value

A list of dosing regimens.

Get a catalog entry

Description

Get a catalog entry

Usage

```
library_get(library_id, root = library_catalog_root())
```

Arguments

library_id	Entry identifier.
root	Catalog root.

Value

List with manifest, paths, and summary.

Persistent LibeRary home

Description

Defaults to Documents/LibeR/library on Windows and ~/LibeR/library on Linux/macOS, alongside the default LibeRation workspace. Set LIBERARY_HOME to override it.

Usage

```
library_home(create = FALSE)
```

Arguments

create	Create the directory.
--------	-----------------------

Value

Normalized directory path.

Build a provider-neutral multimodal user message

Description

Image files are encoded as data URLs for OpenAI-compatible chat endpoints. The Ollama adapter converts the same blocks to its native `images` array.

Usage

```
library_image_message(  
  prompt,  
  image_paths,  
  labels = NULL,  
  detail = c("high", "auto", "low", "original")  
)
```

Arguments

<code>prompt</code>	Text instruction.
<code>image_paths</code>	Local PNG, JPEG, WebP, or GIF paths.
<code>labels</code>	Optional labels inserted before each image, normally page ids.
<code>detail</code>	Image detail hint.

Value

A chat message list accepted by `library_llm_chat()`.

Infer an executable NONMEM implementation from model semantics

Description

Reported ADVAN/TRANS values are retained as reported. Otherwise common linear one-, two-, and three-compartment models are mapped deterministically; ambiguous/general models retain candidates and require review.

Usage

```
library_infer_implementation(  
  structural_model,  
  route = NULL,  
  parameters = list(),  
  dose = NULL  
)
```

Arguments

structural_model	Structural-model object.
route	Administration route.
parameters	Extracted parameter object.
dose	Optional dose description.

Value

Implementation record with provenance, rationale, and alternatives.

Create a typed LibeRties literature job

Description

The resulting job contains data, provider/model choices and evidence only; it cannot contain executable R code. API keys are resolved from environment variables on the worker. Full-text transfer requires explicit confirmation.

Usage

```
library_job(  
  task = c("trriage", "parse", "index", "dual_extract", "assess",  
  "adjudicate"),  
  metadata,  
  extraction = NULL,  
  full_text = "",  
  supplement_text = "",  
  pdf_path = NULL,  
  text_lane = NULL,  
  vision_lane = NULL,  
  comparison = NULL,  
  cfg = NULL,  
  run_assessment = TRUE,  
  confirm_transfer = FALSE,  
  label = NULL  
)
```

Arguments

task	One of triage, parse, index, dual_extract, assess, or adjudicate.
metadata	Publication metadata.
extraction	Proposed extraction for an assessment job.
full_text	Optional article text.

supplement_text	Optional supplement/control-stream text.
pdf_path	Optional PDF for parse, dual extraction, or adjudication.
text_lane	Text-lane result for an adjudication job.
vision_lane	Vision-lane result for an adjudication job.
comparison	Field comparison for an adjudication job.
cfg	LibeRary configuration.
run_assessment	Also assess the extraction produced by an index job.
confirm_transfer	Confirm that supplied text may be sent to the queue worker.
label	Job label.

Value

A `liber_job` accepted by local and remote LibeRties queues.

List catalog entries

Description

List catalog entries

Usage

```
library_list(status = NULL, root = library_catalog_root())
```

Arguments

status	Optional status filter (e.g. "validated", "stub").
--------	--

root	Catalog root; default <code>library_catalog_root()</code> .
------	---

Value

Data frame of summary fields per entry.

Check whether an LLM role is usable

Description

Check whether an LLM role is usable

Usage

```
library_llm_available(cfg = NULL, role = .library_llm_roles())
```

Arguments

cfg
LibeRary configuration.

role
One of triage, indexing, vision, assessment, or adjudication.

Value

TRUE or FALSE.

Send a chat request through a configured LibeRary LLM role

Description

Send a chat request through a configured LibeRary LLM role

Usage

```
library_llm_chat(  
  messages,  
  cfg = NULL,  
  role = .library_llm_roles(),  
  format = NULL,  
  sensitive = TRUE  
)
```

Arguments

messages	List of role/content messages.
cfg	LibeRary configuration.
role	One of triage, indexing, vision, assessment, or adjudication.
format	NULL, "json", or a JSON schema list.
sensitive	Whether the request contains publication or patient-derived content.

Value

A library_llm_response with content, model, provider, usage and timing.

Discover models offered by an LLM endpoint

Description

Discover models offered by an LLM endpoint

Usage

```
library_llm_models(  
  provider = NULL,  
  cfg = NULL,  
  role = .library_llm_roles(),  
  refresh = FALSE  
)
```

Arguments

provider	Provider id from <code>library_llm_providers()</code> .
cfg	LibeRary configuration or NULL.
role	Configuration role, used for endpoint defaults.
refresh	Ignore the short in-memory discovery cache.

Value

A data frame with model ids and endpoint metadata.

Language-model providers available to LibeRary

Description

ollama is local by default. openai uses the official API, while openai_compatible supports local servers and gateways implementing the common models and chat-completions endpoints.

Usage

```
library_llm_providers()
```

Value

A data frame describing the provider registry.

Read model control stream for an entry

Description

Read model control stream for an entry

Usage

```
library_model(library_id, root = library_catalog_root())
```

Arguments

library_id	Entry identifier.
root	Catalog root.

Value

Character vector of `model.ct1` lines.

Enrich a LibeRary extraction with executable semantics

Description

This is backward compatible: legacy population, route, subject-count, structural-model, and parameter fields remain in place.

Usage

```
library_model_enrich(extraction, source_locator = "")
```

Arguments

extraction	Parsed literature extraction.
------------	-------------------------------

source_locator	Optional source locator used for appendix-derived data.
----------------	---

Value

Enriched extraction.

Normalize a reported study population into cohorts and descriptors

Description

The original population text is always retained. Summary-statistic records can coexist, so a paper may report mean/SD and a range for the same variable.

Usage

```
library_population_normalize(  
  population,  
  n_subjects = NULL,  
  source_locator = ""  
)
```

Arguments

population	Reported population text or an existing normalized object.
n_subjects	Optional legacy subject count.
source_locator	Optional page/table/appendix locator.

Value

A normalized population object.

Provenance and audit trail for an entry

Description

Provenance and audit trail for an entry

Usage

```
library_provenance(library_id, root = library_catalog_root())
```

Arguments

library_id	Entry identifier.
root	Catalog root.

Value

List with manifest provenance block and optional extraction files.

Evaluate whether an extracted catalogue model is ready for validation

Description

This deterministic gate keeps machine-extracted entries quarantined until their evidence, mapping, control stream, and executable prediction path are internally consistent. Passing the gate establishes computational qualification; it is not evidence that the publication or model is correct.

Usage

```
library_qualification_check(  
  library_id,  
  root = library_catalog_root(),  
  min_confidence = 0.8,  
  simulate = TRUE  
)
```

Arguments

library_id	Catalogue identifier.
root	Catalogue root.
min_confidence	Minimum overall extraction confidence.
simulate	Run a deterministic, residual-free simulation smoke test.

Value

A library_qualification report.

Score LibeRary predictions against a reference corpus

Description

Strict and silver-tier results are reported separately. Additional values in a silver reference are labelled unverified extras, not hallucinations, because the appendix itself may be incomplete.

Usage

```
library_reference_benchmark(  
  root,  
  predictions,  
  task = c("extraction", "triage"),  
  partition = "test",  
  output_dir = NULL,  
  relative_tolerance = 0.05,  
  absolute_tolerance = 1e-08,  
  probability_threshold = 0.3,  
  prediction_variant = "prediction"  
)
```

Arguments

root	Reference corpus root.
predictions	Directory produced by <code>library_reference_run()</code> or an equivalent runner.
task	"extraction" or "triage".
partition	Partition to score.
output_dir	Optional report directory.

relative_tolerance

Relative numeric tolerance.

absolute_tolerance

Absolute numeric tolerance.

probability_threshold

First-pass threshold for triage sensitivity/specificity.

prediction_variant

Score the primary prediction or a stored text, vision, or reconciled dual-pipeline variant.

Value

Benchmark summary and per-record results.

Build a versioned AED PK/PD reference corpus

Description

Converts the v2 systematic-review supplement tables into a separate, auditable LibeRary-style JSON corpus. Appendix text is retained verbatim and normalized fields are marked as silver-tier until checked against the source publication. Article-level deterministic partitions prevent the same PMID from leaking between training, validation, and test data.

Usage

```
library_reference_build(  
  source_dir,  
  output_dir,  
  version = "0.1.0",  
  partition_seed = 20260716L,  
  validation_fraction = 0.15,  
  test_fraction = 0.15  
)
```

Arguments

source_dir	Directory containing Supplements and the review files.
output_dir	New directory for this immutable corpus version.
version	Corpus version. Use a new directory for every version.
partition_seed	Stable string or integer used for article partitions.
validation_fraction	Fraction assigned to validation.
test_fraction	Fraction assigned to the locked test set.

Value

Corpus manifest invisibly.

Compare a systematic-review reference with a LibeRary extraction

Description

Returns the historical appendix transcription, the normalized reference target, a selected prediction variant, and a leaf-level comparison table. The function is read-only; curation is applied only through `library_reference_revise()`.

Usage

```
library_reference_compare(  
  reference_id,  
  root,  
  predictions,  
  variant = "prediction",  
  relative_tolerance = 0.05,  
  absolute_tolerance = 1e-08  
)
```

Arguments

<code>reference_id</code>	Stable model identifier in the reference corpus.
<code>root</code>	Reference-corpus root.
<code>predictions</code>	Prediction directory produced by <code>library_reference_run()</code> (or its <code>predictions</code> subdirectory).
<code>variant</code>	Primary prediction or a stored extraction variant such as <code>text</code> , <code>vision</code> , or <code>reconciled</code> .
<code>relative_tolerance</code>	Relative numeric tolerance used to label matches.

`absolute_tolerance`

Absolute numeric tolerance used to label matches.

Value

A list containing the record, prediction envelope, available variants, selected prediction, and comparison data frame.

Read a reference-corpus model

Description

Read a reference-corpus model

Usage

```
library_reference_get(reference_id, root)
```

Arguments

reference_id	Stable reference id.
root	Reference corpus root.

Value

Parsed model record.

List reference-corpus models

Description

List reference-corpus models

Usage

```
library_reference_list(root, partition = NULL, tiers = NULL)
```

Arguments

root	Reference corpus root.
partition	Optional train, validation, or test filter.
tiers	Optional quality-tier filter.

Value

Data frame with one row per model.

Create a reviewed successor version of a reference corpus

Description

The source version is never edited. Curation decisions are applied to a new directory and recorded in each model's audit trail. Test records can be promoted for strict scoring but can never become training eligible.

Usage

```
library_reference_revise(  
  root,  
  output_dir,  
  decisions,  
  version,  
  curator,  
  corrections = NULL  
)
```

Arguments

root	Existing reference corpus root.
output_dir	Empty directory for the successor version.
decisions	Data frame or CSV path with reference_id, tier, and optional training_eligible, notes, and review_status columns.
version	New semantic corpus version.
curator	Name or identifier of the reviewer.
corrections	Optional data frame or CSV path containing field-level corrections with reference_id, JSON pointer, decision source, and value_json. Corrections are applied only to the successor's normalized

reference\$extraction_target; the historical raw transcription remains untouched.

Value

New manifest invisibly.

Run the current LibeRary pipeline on a protected reference partition

Description

Only article metadata and source documents are supplied to the model. The appendix-derived target is never placed in a prompt or copied into the run directory. Results are resumable and can be scored separately with `library_reference_benchmark()`.

Usage

```
library_reference_run(  
  root,  
  source_dir,  
  output_dir,  
  cfg = NULL,  
  task = c("extraction", "trriage"),  
  partition = "test",  
  extraction_mode = c("dual", "text"),  
  limit = Inf,  
  resume = TRUE,  
  assess = FALSE,  
  progress = NULL  
)
```

Arguments

root	Reference corpus root.
source_dir	Original AED PK/PD review directory containing the PDFs.
output_dir	Directory for prediction envelopes.
cfg	LibeRary configuration.

task	"extraction" or "triage".
partition	Protected partition to run.
extraction_mode	For extraction, use the complete independent text/vision/adjudication pipeline or the legacy text-only lane.
limit	Optional maximum number of unique articles.
resume	Reuse completed prediction files.
assess	Run LibeRary's independent assessment after extraction.
progress	Optional callback accepting fraction, message, index, total.

Value

Run manifest invisibly.

Launch the LibeRary reference comparison and curation application

Description

Displays the historical systematic-review appendix, normalized reference, and a selected LibeRary extraction side by side. Field decisions are kept in session until the reviewer explicitly creates an immutable successor corpus.

Usage

```
library_reference_shiny(  
  corpus = NULL,  
  predictions = NULL,  
  source_dir = NULL,  
  host = "127.0.0.1",  
  port = NULL,  
  launch.browser = TRUE  
)
```

Arguments

corpus	Reference-corpus root. When NULL, the application tries LIBRARY_REFERENCE_ROOT and the local validation directory.
predictions	Prediction directory produced by library_reference_run().
source_dir	Original systematic-review directory used to locate PDFs.
host	Passed to shiny::runApp().
port	Port or NULL.

`launch.browser`

Open a browser.

Value

Invisibly, the Shiny app object.

Export leakage-checked adapter-training examples

Description

Extraction examples pair source-paper text with reviewed JSON targets. The appendix text itself is never included in the user message. By default only reviewed A/B records explicitly marked training-eligible are exported. Locked test data is rejected unconditionally.

Usage

```
library_reference_training_export(  
  root,  
  source_dir,  
  output_dir,  
  tasks = c("extraction", "triage"),  
  partitions = c("train", "validation"),  
  tiers = c("A", "B"),  
  allow_silver = FALSE,  
  require_training_eligible = TRUE,  
  max_source_chars = 60000L  
)
```

Arguments

root	Reference corpus root.
source_dir	Original AED PK/PD review directory containing PDFs.
output_dir	Empty or existing output directory.
tasks	Any of "extraction" and "triage".
partitions	Training/development partitions; test is forbidden.

tiers

Eligible quality tiers.

allow_silver

Permit C/D extraction targets for experiments. These are labelled silver and should not be used for a production adapter.

require_training_eligible

Require the model-level review flag.

max_source_chars

Maximum source-paper characters per extraction example.

Value

Export manifest invisibly.

Locate the optional local-adapter training tools

Description

Locate the optional local-adapter training tools

Usage

```
library_reference_training_files()
```

Value

Named paths to the Python trainer and its requirements file.

Validate a reference corpus and its leakage safeguards

Description

Validate a reference corpus and its leakage safeguards

Usage

```
library_reference_validate(root, check_hashes = FALSE, source_dir = NULL)
```

Arguments

root	Reference corpus root.
check_hashes	Recompute locally available appendix/PDF hashes when source_dir is supplied.
source_dir	Optional original AED PK/PD review directory.

Value

A list containing valid, errors, warnings, and counts.

Permanently empty the mutable LibeRary repository

Description

Removes all LibeRary-managed catalogue entries, acquired publications, parsed documents, manifests, triage records, caches, and job logs beneath `data_dir`. The upgrade-safe configuration YAML and any unrelated files in the data-directory root are retained. Package-bundled example entries are not copied back into a repository explicitly emptied by this function.

Usage

```
library_repository_wipe(confirmation, data_dir = library_home())
```

Arguments

<code>confirmation</code>	Must be exactly "YES" (case-sensitive).
---------------------------	---

<code>data_dir</code>	Persistent LibeRary data directory.
-----------------------	-------------------------------------

Value

A summary containing the normalized root and number of removed filesystem entries, invisibly.

Prepare an auditable article-reproduction plan

Description

A plan is conservative by default: generated parameter defaults block execution unless explicitly allowed. Missing concentration targets do not prevent simulation, but they do prevent quantitative reproduction scoring.

Usage

```
library_reproduction_plan(  
  extraction,  
  times = NULL,  
  n_subjects = NULL,  
  allow_generated_defaults = FALSE  
)
```

Arguments

extraction	Enriched or legacy LibeRary extraction.
times	Optional simulation times.
n_subjects	Optional synthetic-population size.
allow_generated_defaults	Permit generated model parameters.

Value

A `library_reproduction_plan`.

Execute an article-reproduction plan with LibeRation

Description

Execute an article-reproduction plan with LibeRation

Usage

```
library_reproduction_run(  
  x,  
  output_dir = NULL,  
  nsim = 200L,  
  seed = 20260716L,  
  n_cores = 1L,  
  allow_generated_defaults = FALSE  
)
```

Arguments

x	Extraction or object returned by <code>library_reproduction_plan()</code> .
output_dir	Optional artifact directory.
nsim	Number of population simulations.
seed	RNG seed.
n_cores	Simulation workers.
allow_generated_defaults	Allow generated parameter defaults.

Value

A library_reproduction_result.

Compare simulated and digitized concentration-time targets

Description

Compare simulated and digitized concentration-time targets

Usage

```
library_reproduction_score(simulated, targets, value = "IPRED")
```

Arguments

simulated	Simulation output from LibeRation.
targets	Target data frame from a reproduction plan.
value	Simulation column, normally IPRED or DV.

Value

Per-point comparison and summary metrics.

Record a human review decision

Description

Status changes are auditable. Promotion to `validated` requires an explicit confirmation when any model code or parameter value was generated.

Usage

```
library_review(  
  library_id,  
  status = c("review", "validated", "deprecated"),  
  reviewer,  
  notes = "",  
  confirm_generated = FALSE,  
  root = library_catalog_root()  
)
```

Arguments

library_id	Catalogue id.
status	New workflow status.
reviewer	Reviewer name or identifier.
notes	Review notes.
confirm_generated	Confirm that generated suggestions were reviewed.
root	Catalogue root.

Value

Updated manifest, invisibly.

Save upgrade-safe LibeRary settings

Description

API keys are removed before writing; configure their environment-variable names in YAML and put the actual secrets in the worker environment.

Usage

```
library_save_config(cfg, path = library_config_path(create = TRUE))
```

Arguments

cfg	LibeRary configuration.
-----	-------------------------

path	Destination YAML path.
------	------------------------

Value

Path, invisibly.

Search the catalog

Description

Keyword search over title, compound, population, keywords, and abstract (when present in manifest).

Usage

```
library_search(  
  query = "",  
  compound = NULL,  
  population = NULL,  
  status = NULL,  
  min_confidence = NULL,  
  advan = NULL,  
  model_type = NULL,  
  assessment = NULL,  
  root = library_catalog_root()  
)
```

Arguments

query	Free-text query (case-insensitive).
compound	Filter by compound name (partial match).
population	Filter by population (partial match).
status	Filter by status.
min_confidence	Minimum overall confidence score.
advan	Optional ADVAN number.

<code>model_type</code>	Optional model type (pk, pd, or pkpd).
-------------------------	--

<code>assessment</code>	Optional assessment verdict.
-------------------------	------------------------------

<code>root</code>	Catalog root.
-------------------	---------------

Value

Data frame of matching entries (same columns as `library_list()`).

Launch the LibeRary catalog browser (Shiny)

Description

Browse pharmacometric models in the catalog: search, filter, inspect manifests, preview control streams, and import entries into a LibeRation workspace.

Usage

```
library_shiny(  
  catalog = NULL,  
  mode = c("catalog", "ingest", "reference"),  
  host = "127.0.0.1",  
  port = NULL,  
  launch.browser = TRUE,  
  reference_root = NULL,  
  predictions = NULL,  
  source_dir = NULL  
)
```

Arguments

catalog	Root directory of the catalog (see <code>library_catalog_root()</code>).
mode	Open the catalogue browser, literature-ingestion workflow, or reference comparison and curation workflow.
host	Passed to runApp.
port	Port; NULL picks a random free port.
launch.browser	Open browser when TRUE.

reference_root	Optional reference-corpus root used in reference mode.
predictions	Optional benchmark prediction directory used in reference mode.
source_dir	Optional source-review directory used to open publication PDFs in reference mode.

Details

Requires Suggested packages shiny and DT.

Value

Invisibly, the Shiny app object.

Locate the source PDF for a catalogue entry

Description

Resolves the immutable document bundle recorded in the entry provenance and falls back to the persistent ingest inbox for older entries.

Usage

```
library_source_pdf(library_id, root = library_catalog_root())
```

Arguments

library_id	Entry identifier.
------------	-------------------

root	Catalog root.
------	---------------

Value

Normalized PDF path, or an empty string when no source PDF is available.

Import a catalogue model into a LibeRation workspace

Description

The control stream is parsed and compiled by LibeRation before it is stored. Catalogue identity, version, evidence provenance, and qualification state are retained in the immutable model-version provenance record.

Usage

```
library_use_in_workspace(  
  library_id,  
  project = NULL,  
  workspace = NULL,  
  version_label = NULL,  
  root = library_catalog_root()  
)
```

Arguments

library_id	Catalogue entry id.
project	Existing project id or a name for a new project. By default a stable project id is derived from <code>library_id</code> .
workspace	An <code>nm_workspace</code> , path, or <code>NULL</code> for LibeRation's default.
version_label	Optional model-version label.
root	Catalogue root.

Value

Project id, model-version id, workspace, and compatibility report.

Validate a LibeRary manifest or catalogue entry

Description

Validate a LibeRary manifest or catalogue entry

Usage

```
library_validate(  
  library_id = NULL,  
  manifest = NULL,  
  root = library_catalog_root(),  
  check_artifact = TRUE  
)
```

Arguments

library_id	Optional catalogue id.
manifest	Optional parsed manifest.
root	Catalogue root.
check_artifact	Verify that the model artifact exists.

Value

A list of valid, errors, and warnings.

Execute a typed LibeRary worker task

Description

This is the only LibeRary entry point callable by LibeRties workers.

Usage

```
library_worker_task(type, payload, arguments = list())
```

Arguments

type	A typed LibeRary pipeline job.
------	--------------------------------

payload	Validated data payload.
---------	-------------------------

arguments	Named task controls.
-----------	----------------------

Value

Extraction or assessment result.